

NAPF, Switzerland

WMO: 066390

Lat: 47.0047N

Lon: 7.9400E

Elev: 1406

StdP: 85.54

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-12.6	-10.7	-20.7	0.7	-2.5	-17.2	1.0	-3.1	15.1	1.3	13.6	1.1	2.7	10	0.652

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.4	23.3	16.3	21.7	15.6	20.3	14.9	17.6	21.8	16.5	20.5	15.5	19.3	2.3	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
15.9	13.4	20.0	14.9	12.6	18.7	14.0	11.8	17.7	55.4	22.0	51.8	20.5	48.7	19.3	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.7	11.0	9.4	DB	-14.3	25.5	3.1	1.7	-16.5	26.7	-18.3	27.7	-20.0	28.7	-22.2	29.9	(4)
(5)			WB	-14.8	19.7	3.0	1.1	-16.9	20.5	-18.7	21.1	-20.3	21.8	-22.5	22.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.9	-1.5	-1.6	1.1	4.4	8.6	12.4	14.2	14.0	10.0	7.2	2.2	-0.4		
	DBStd	7.27	4.46	5.15	4.54	4.74	4.30	4.53	3.91	4.03	3.95	4.45	4.91	4.71		
	HDD10.0	1974	358	326	277	178	79	26	8	9	49	106	234	324		
	HDD18.3	4557	616	559	535	419	303	184	138	144	249	344	483	582		
	CDD10.0	484	0	0	0	8	35	98	138	133	49	20	2	0		
	CDD18.3	25	0	0	0	0	1	6	9	9	0	0	0	0		
	CDH23.3	47	0	0	0	0	0	12	18	17	0	0	0	0		
	CDH26.7	2	0	0	0	0	0	1	1	1	0	0	0	0		
(14) Wind	WSAvg	3.9	4.7	4.5	4.3	3.4	3.4	3.1	3.3	3.2	3.4	3.8	4.3	5.1		
(15) Precipitation	PrecAvg	1288	80	79	84	98	136	136	144	133	104	98	96	100		
	PrecMax	1615	202	196	258	200	230	244	291	252	181	174	211	217		
	PrecMin	963	14	27	34	21	38	59	46	22	48	8	4	0		
	PrecStd	177	49	45	50	48	55	46	55	59	41	41	50	54		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.4	10.8	12.1	17.3	20.9	25.0	25.4	25.3	19.9	17.7	14.0	10.5		
		MCWB	2.8	2.8	6.3	10.8	14.1	17.1	17.1	16.9	15.0	10.2	6.6	2.9		
	2%	DB	7.6	8.6	10.3	14.6	18.4	22.6	23.1	23.1	18.2	16.0	12.2	8.8		
		MCWB	1.8	2.1	5.2	9.0	12.6	16.1	16.3	16.0	13.8	10.1	6.1	2.4		
	5%	DB	5.8	6.8	8.9	12.7	16.6	20.7	21.4	21.5	16.9	14.4	10.7	7.0		
		MCWB	1.3	1.4	4.2	7.7	11.3	15.1	15.5	15.5	13.0	9.5	5.4	1.6		
	10%	DB	4.3	5.0	7.4	11.0	14.8	19.0	19.8	19.8	15.6	13.1	9.0	5.5		
		MCWB	0.9	1.1	3.5	6.5	10.4	14.1	14.7	14.9	12.1	9.0	5.0	1.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.1	4.7	7.5	11.4	15.3	19.0	18.7	18.8	16.2	13.0	8.4	5.4		
		MCDB	6.4	7.3	10.5	16.0	19.6	23.2	23.2	22.8	18.7	15.4	11.1	6.8		
	2%	WB	3.7	3.4	6.2	9.6	13.4	17.1	17.3	17.4	14.8	11.5	7.4	4.2		
		MCDB	5.5	5.9	9.0	13.7	17.4	21.4	21.7	21.6	17.3	13.8	10.1	6.3		
	5%	WB	2.5	2.5	5.1	8.3	12.0	15.8	16.2	16.2	13.6	10.5	6.4	3.3		
		MCDB	4.6	5.3	7.8	12.0	15.7	19.6	20.3	20.0	16.2	12.8	9.2	5.4		
	10%	WB	1.4	1.6	4.1	7.1	10.8	14.6	15.3	15.3	12.6	9.6	5.5	2.4		
		MCDB	3.7	4.5	6.8	10.5	14.0	18.1	19.1	19.0	15.0	12.2	8.4	4.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.3	4.4	4.8	5.7	6.3	6.5	6.4	5.8	4.9	4.8	4.3	4.3		
		MCDBR	5.7	6.0	6.1	7.5	7.8	7.7	7.8	7.0	6.3	5.9	6.0	5.8		
	5% WB	MCWBR	3.2	3.5	4.2	5.4	5.8	6.1	5.5	5.4	4.9	4.4	3.4	3.2		
		MCWBR	5.0	5.4	5.6	7.2	7.5	7.4	7.3	6.7	5.8	5.1	5.3	5.1		
(40) Clear-Sky Solar Irradiance	taub		0.252	0.274	0.308	0.339	0.352	0.356	0.357	0.348	0.327	0.306	0.276	0.250		
	taud		2.454	2.410	2.395	2.359	2.364	2.387	2.371	2.409	2.458	2.508	2.522	2.467		
	Ebn at Noon		864	907	919	916	912	907	901	896	887	858	826	825		
	Edh at Noon		64	83	100	114	118	116	116	107	93	76	61	57		
(44) All-Sky Solar Radiation	RadAvg		1.29	2.16	3.35	4.37	4.96	5.82	5.68	4.94	3.79	2.43	1.37	1.06		
	RadStd		0.15	0.30	0.48	0.65	0.48	0.54	0.50	0.47	0.36	0.26	0.18	0.17		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	+0.99	+0.57	+0.31	N/A	N/A	N/A	+9
(47) Regional (120 neighbors)		N/A	N/A	N/A	+0.74	+0.43	+0.33	N/A	N/A	+57	+15

Nomenclature: See separate page